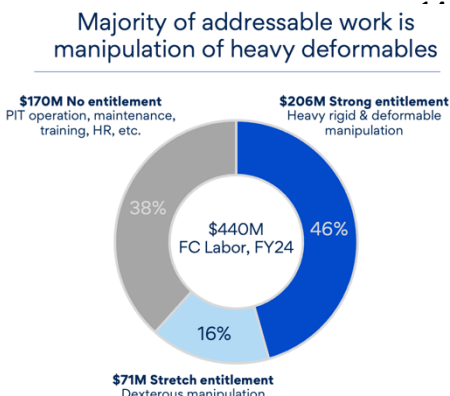


Physical AI 2026 OP1

Executive Summary

Chewy's Physical AI strategy is built on a simple premise: data will determine how robotics companies shape the capabilities of the coming generation of humanoids. Just as language models learned to reason from vast corpora of text, physical AI models learn to act by training on data of robots performing real-world tasks. Today, that data are scarce. By publishing high-quality datasets of robots manipulating deformable items—the same tasks that dominate our fulfillment centers—we make Chewy's work the curriculum that trains the industry. Robots built by others will arrive already skilled in the problems we care about most. **This is our first strategic lever: influence through dataset publication.** By defining the warehouse tasks and success metrics that future physical AI models must satisfy, we shape the evaluation standards for the industry itself, guiding vendors to measure their progress on the work that matters to Chewy. **Our second lever is readiness through orchestration:** architecting the Robot Management System (RMS) that will coordinate a mixed fleet of humanoids inside Chewy's fulfillment centers once the hardware matures.



Humanoids represent one of the largest long-term automation opportunities in Chewy's network. Of the roughly **\$440 million in FC labor costs in 2024**, more than **\$200 million** went to heavy rigid and deformable object handling—the kind of physical work that humanoids could take on if they acquire the right capabilities. Our plan is to help the industry get there. The TREAT Lab is built to replicate the workflows that matter most—picking, packing, and truck loading, which together account for about **\$152 million** in annual spend. By capturing the data, motion patterns, and success metrics of these high-value tasks, we create the datasets and evaluation standards that will guide the development of humanoids toward the capabilities Chewy needs most.

The technology behind humanoid robots is advancing quickly but remains several years from broad commercial deployment. Chewy is using the intervening period to shape how the field tackles the manipulation challenges that define our operations while building readiness for adoption at scale. The commissioned **TREAT¹ Lab** at BOS3 serves as foundational infrastructure for this plan, designed to operate with increasing continuity as our systems mature. **Like a well-run factory, it is built for round-the-clock production—a North Star that stretches us toward more robots, more capable orchestration, and richer data generation.** Multiple robots from different vendors will work in parallel under warehouse-realistic conditions. We will begin with teleoperation to generate high-quality demonstrations of deformable manipulation and coordinated handling. As AI capabilities advance, the same workflows transition to semi-autonomous and then autonomous operation, which increases both data volume and task complexity while preserving continuity of evaluation.

Coordinating an expanding and diverse fleet requires an orchestration layer that assigns, monitors, and optimizes robotic work with the same rigor that a warehouse management system applies to human labor. The RMS fills this role. It interprets fulfillment workflows, dispatches tasks according to each robot's unique strengths and limitations, and ensures that heterogeneous platforms collaborate safely and productively. As more vendors test in the TREAT Lab, we gain direct visibility into their APIs and control approaches, which strengthens our ability to allocate work and optimize performance in a mixed-vendor environment. The RMS becomes the control system for the data factory and the basis for next-generation warehouse operations.

The result is a self-reinforcing system: testing robots defines the warehouse capabilities we care about; those capabilities generate valuable datasets that reflect Chewy's operational reality; the datasets attract vendors and researchers who seek validation on the tasks that matter; and every new participant expands the data factory, improves the RMS, and accelerates learning. **Run the lab, grow the fleet, publish the data, raise the standard, attract more robots, repeat.** Through the TREAT Lab, Chewy sets the benchmark by which warehouse humanoids are judged and meets industry thinkers where they work through peer-reviewed papers and open artifacts.

¹ Testing and Research of Embodied AI Technologies.

In 2026 we move from preparation to execution across three frontiers that reinforce the flywheel. First, we expand data quality with tactile sensing so models learn contact-rich manipulation of heavy, flexible bags. Second, we raise simulation fidelity, validate digital twins against the TREAT Lab, and evaluate World Foundation Models to scale synthetic data where it adds value. Third, we progress the RMS from architecture to algorithms, proving heterogeneous fleet coordination in simulation and then in the lab. **By year-end, we aim to show datasets that improve deformable manipulation performance, simulation assets that shorten controller bring-up, and an end-to-end RMS demonstration with multiple humanoid platforms working together.** These investments ensure that when humanoids reach economic readiness, they arrive pretrained on Chewy's work, validated by Chewy's standards, and orchestrated by Chewy's systems.

Tenets

Our Physical AI program is guided by a set of durable principles that remain constant even as our technical plans evolve. We have refined the fleet-scale tenet to emphasize heterogeneity, recognizing that we must build for vendor optionality.

1. **Train the World's Robots on Chewy Work.**

The fastest way to influence humanoid roadmaps is to make Chewy's work their training ground. By publishing open, high-quality datasets of robots manipulating deformable objects, we shape the embodied AI models that vendors will bring to market. Every robot we evaluate should arrive with skills already tuned to our needs.

2. **Partner Broadly, Stay Top of Mind.**

We avoid exclusivity. By engaging multiple vendors, academics, and component suppliers, we ensure Chewy remains top of mind when design decisions are made. Non-exclusive collaboration positions us to pivot as the field evolves.

3. **Adopt When Ready, Not When Flashy.**

Humanoids enter our FCs only when economics prove out. The bar is explicit: fully loaded cost must undercut human benchmarks with a credible path to savings over time. Novelty is not enough; measurable business outcomes are the standard.

4. **Lead with Technical Credibility.**

We build influence through visible expertise in reinforcement learning, simulation, and orchestration algorithms. Technical credibility is earned through peer-reviewed publications, reproducible datasets, and open artifacts.

5. **Build for Fleet-Scale from Day One.**

We assume a *heterogeneous* fleet, not a monolithic deployment from a single vendor. Our RMS must orchestrate across mobility, dexterity, payload, and API differences. From the outset we design for scale: modular components, clean interfaces, and fleet-level intelligence that compounds as robots learn together.

2025 Highlights and Lowlights

2025 was a foundational year for Chewy's Physical AI program. We built the physical and conceptual infrastructure that now anchors our long-term strategy, learning as much from our early missteps as from our progress.

Highlights. The year began with an empty warehouse and ended with the TREAT Lab, a fully instrumented test facility replicating fulfillment center operations, including a full zone of a Pick Module. The lab now includes FC infrastructure, humanoid and custom-built robots, fabrication equipment, instrumentation and data-collection software capable of benchmarking humans and humanoids under identical conditions. We also developed high-fidelity digital twins of these environments and commissioned a Chewy-designed, two-armed robot capable of lifting heavy deformable objects, establishing the foundation for our dataset publication efforts.

Lowlights. Our early outreach to robot manufacturers revealed the pitfalls of engaging too narrowly and too soon. Initial conversations tied testing access to purchase commitments and exclusivity agreements, with vendors positioning themselves for early sales rather than shared research. At the time, our own strategy was still forming, and the lack of clarity allowed the discussions to be framed as transactional rather than scientific. Recognizing that coupling technological progress to economic incentives risked premature vendor lock-in, we reframed our approach entirely. This experience led us to the model we pursue today: open publication of high-quality training data that drives progress across the industry without binding us to any one manufacturer. Importantly, the success of this strategy does not depend on competing for vendors' attention or committing to purchases before the technology is ready. We will adopt humanoids only when the market matures and a vendor's roadmap and partnership model align with Chewy's long-term vision.

We also initially underestimated the challenge of accurately modeling the physical properties of deformable objects, particularly the heavy, flexible bags that define much of Chewy’s fulfillment work. Our 2025 efforts to simulate these materials in NVIDIA Omniverse revealed the limits of existing physics engines. While visually convincing, the models behaved unrealistically under stress, with particles sometimes passing through bag surfaces or collapsing in nonphysical ways. For a time, this raised real concern that the industry lacked a viable path to high-fidelity modeling of heavy deformables and that Chewy alone could not solve it. Shortly afterward, NVIDIA announced Newton, a higher-fidelity soft-body simulator that addresses many of these shortcomings. Its release gave us a renewed path forward, positioning Chewy to be an early adopter of improved simulation tools and to focus our modeling on heavy deformable objects that produce training data directly relevant to the tasks we aim to automate.

2026 Plan

In 2026, Chewy will **operate the TREAT Lab as a data factory**, a long-horizon effort aimed at continuous operation. Achieving uninterrupted activity is not the immediate goal, but it serves as the guiding principle that shapes our systems, partnerships, and research investments. The ambition to reach round-the-clock production gives structure to every component of our Physical AI strategy. The more consistently the lab runs, the faster the flywheel turns—robots generate data, data attract partners, partners expand capability, and capability accelerates readiness for humanoid adoption.

Our work continues to advance along two connected levers. The first is influence through dataset publication, ensuring that the embodied AI models powering humanoids are trained on Chewy’s work. The second is readiness through orchestration, developing the Robot Management System (RMS) that coordinates the lab’s growing fleet and will ultimately govern robots inside our fulfillment centers. Together, they form a closed loop where each experiment strengthens Chewy’s position as both a data producer and a future operator of mixed-vendor humanoid fleets.

Factory Lines of 2026

Each 2026 research theme functions as a line on the production floor—tactile data, simulation, and orchestration—working in parallel but connected through the RMS. These lines convert robotic motion into structured, high-value datasets that expand Chewy’s influence and readiness.

Line 1: The Tactile Data Line. This line develops the sense of touch required for humanoids to handle the heavy, flexible items that dominate Chewy’s fulfillment work. The TREAT Lab will conduct repeated, instrumented trials where robots and humans perform identical manipulation tasks, producing synchronized visual, proprioceptive, and tactile data. In partnership with Analog Devices, we will evaluate new tactile sensors that capture detailed contact patterns during grasping and lifting. These datasets reveal the mechanical truths of deformable manipulation and become part of the training material that vendors and researchers rely on. As the tactile line matures, every improvement in sensing precision amplifies the data factory’s value: richer data produce stronger models, which make the lab’s robots more capable, which in turn draw more partners into our orbit.

Line 2: The Simulation Line. The simulation line extends the factory into the digital domain. In 2026, we will expand our library of deformable-object models, validate digital twins of TREAT Lab environments, and integrate emerging World Foundation Models (WFMs) to generate physics-aware episodes of robot behavior. Simulation allows the lab to operate in two dimensions: physical experiments that measure real performance and digital ones that explore new configurations at scale. This blended approach accelerates learning while maintaining a focus on realism. By publishing calibrated deformable assets and a Synthetic-Data Playbook, Chewy will establish the standard for combining synthetic and physical data in embodied AI. The simulation line gives the factory reach far beyond its physical limits and ensures that every hour of lab time multiplies into many hours of virtual training.

Line 3: The Orchestration Line. The orchestration line builds the RMS as the factory’s control system. As more robots from diverse vendors operate under one roof, coordination becomes essential. The RMS translates fulfillment workflows into robotic instructions, assigns work based on each platform’s strengths, and monitors fleet performance for safety and efficiency. In 2026, we will progress from architecture to algorithms, focusing on large-scale task assignment, routing, and collaborative localization. Each experiment refines the RMS and produces the operational data needed to optimize throughput across heterogeneous fleets. The more robots that enter the lab, the better the orchestration becomes, completing the feedback loop that turns experimentation into infrastructure.

The year closes with a unified outcome: a data factory that operates across tactile, simulated, and orchestrated dimensions. The TREAT Lab serves as the production floor, the RMS as the control system, and the data as the product. Each turn of this flywheel compounds Chewy’s influence, technical credibility, and future readiness. When humanoids reach economic viability, they will be trained on Chewy’s work, validated by Chewy’s standards, and orchestrated by Chewy’s systems.

Deliverables and Guardrails

In 2026 we shift from preparation to execution. The TREAT Lab, simulation assets, and orchestration architecture built in 2025 now enable controlled experimentation and measurable progress. For each research theme, improvement targets will be assessed against baselines that are established concurrently. In practice, this means we will develop reference models and benchmarks alongside new techniques (for example, creating a baseline deformable-grasping model that omits tactile data while evaluating tactile-enhanced models under the same conditions). These parallel baselines allow us to quantify the incremental value of each new modality, algorithm, or simulation technique as the year unfolds.

The Tactile Data Line. We will publish open datasets of humans and humanoids performing core warehouse tasks and fine-tuned models that prove their utility. We will also release a tactile-augmented dataset of two-handed deformable manipulation. Baseline models trained without tactile inputs will be developed in parallel to establish reference performance for deformable grasping. If tactile sensing yields less than 20% improvement over this baseline, we will pause scaling to refine data quality and calibration or will narrow its use to SKU classes where gains are measurable.

The Simulation Line. We will expand our *Deformables Asset Library*, validate digital twins of robots and FC environments, and publish a *Synthetic-Data Playbook* showing how World Foundation Models complement curated datasets. Baseline performance will be measured by comparing models trained solely on existing simulation data with those trained on new deformables assets and WFM-generated content. Our goal is to demonstrate that behaviors learned in simulation also succeed in the physical world—for example, that a robot trained virtually can perform deformable object packing or lifting tasks with a similar rate of success in controlled experiments. If new simulations cannot close this gap within 50% of physical-world performance, we will escalate through external consultants and academic partners. If adding WFM-generated data improves success rates in lab trials by at least 20% compared with models trained only on curated simulation data, we will expand their use; if the gain is smaller, WFMs will remain a supplemental data source.

The Orchestration Line. We will release an RMS architecture with requirements and APIs, publish a prototype algorithm from the architecture, and demonstrate an end-to-end heterogeneous fleet in the TREAT Lab. Baseline orchestration performance will be established using single-agent or homogeneous-fleet simulations, providing reference throughput and latency metrics for heterogeneous RMS algorithms. If the initial task assignment algorithm cannot compute solutions fast enough for real-time re-tasking, we will pivot to approximate methods rather than over-invest in intractable optimization.

Metrics

Our 2026 metrics align with the two levers of Chewy’s Physical AI strategy: influence through dataset publication and readiness through orchestration. Each top-line objective represents proof of progress toward these goals, supported by measurable key performance indicators (KPIs).

Objective 1: Deformable object picking with a tactile-augmented Physical AI model

We will demonstrate that models trained in simulation and fine-tuned on tactile-augmented Chewy datasets can perform deformable object picking tasks in physical trials. Success is defined as consistent, repeatable picking performance in the TREAT Lab that surpasses a baseline model trained without tactile data.

Key Performance Indicators:

- Baseline model trained in simulation using visual and joint angle inputs, achieving at least 500 successful picks in 1,000 controlled trials.
- Tactile-enhanced model fine-tuned on Chewy datasets, achieving $\geq 20\%$ improvement in successful picks relative to the baseline.

- Validation that performance improvements hold across both simulation and physical experiments, confirming transfer from virtual to real settings.
- Publication of results and release of open datasets and models documenting the tactile contribution.

Objective 2: End-to-end heterogeneous fleet demonstration in the TREAT Lab

We will validate that Chewy’s Robot Management System (RMS) can coordinate multiple humanoid platforms under a common orchestration layer. Success is defined as a live demonstration of at least two humanoids from different vendors completing shared fulfillment tasks in the TREAT Lab.

Key Performance Indicators:

- RMS architecture finalized with defined APIs and data flow among subsystems.
- Prototype RMS algorithm (e.g., task assignment or routing) validated in simulation, achieving $\geq 90\%$ throughput improvement over a single-agent baseline.
- RMS algorithms deployed on physical robots, sustaining re-planning latency ≤ 200 ms for a fleet of 100 agents.
- Public release of the first Chewy benchmark for large-scale task assignment or multi-agent path finding, with external citations or downloads from academic and industry partners.

Risks and Mitigations

Immaturity of humanoid autonomy. No humanoid available today has demonstrated reliable long-duration autonomy in unstructured environments. The timeline for maturity remains uncertain. We mitigate by centering our strategy on datasets and orchestration. Progress in autonomy accelerates our plans, but our impact does not depend on it.

Vendor constraints and lock-in. Some vendors may resist open-sourcing data collected from their robots, or attempt to tie access to exclusivity. We mitigate by continuing to develop our own experimental platforms (e.g., BUSTER) and maintaining multiple vendor relationships. This hedge ensures we always retain access to raw robot data.

Simulation fidelity. Inaccurate models of deformable objects could prevent transfer of policies from simulation to the physical world. To mitigate, we will validate digital twins against high-quality experimental data, use varied simulated conditions to build robustness, and partner with outside experts where needed.

Platform dependency. Our simulation stack is evolving, and engine changes can introduce delays. We mitigate by maintaining fallback environments for small-scale experiments and by scoping fidelity only to what the task requires, not perfection.

Tactile sensing dependency. Our tactile program is currently centered on Analog Devices as the sensor supplier. If their roadmap slips, we risk schedule delays. We mitigate by engaging secondary partners such as Tacta Systems and Psyonic and evaluating emerging open-source tactile sensors. We also maintain vision-only baselines so that experiments produce useful data even if tactile integration falters.

World Foundation Models (WFMs). WFMs are both an opportunity and a threat. If the robotics field converges on WFM-generated synthetic data as sufficient, the strategic value of curated experimental datasets may erode. Conversely, if we leverage WFMs effectively, they can multiply the reach of our data and tools. We mitigate by benchmarking WFM-only versus WFM-plus-Chewy pipelines and publishing a short Synthetic-Data Playbook to document explicit use cases. This ensures we remain relevant even if the field shifts toward synthetic-first training.

Organizational coupling risk. The RMS requires clean integration with Chewy’s WMS and CMMS. Excessive modularity may create performance drag, while over-integration risks brittleness. We mitigate by designing for a “middle ground”: modular components with defined couplings where performance demands.

Milestones

Each milestone unlocks the next; Q4 shows externally visible proofs.

Quarter	Line	Milestone	Unlock
Q1 2026	TACTILE	Six-hour teleoperated humanoid trial in PickMod	Validates robot reliability and two-handed operation; establishes first sustained production run in the data factory and baseline for throughput metrics
	TACTILE	Initial tactile instrumentation integrated and tested on humanoid manipulator	Establishes synchronized tactile, vision, and proprioceptive data capture
	SIM	Fine-tune open-weight physical AI model in Chewy's simulated environment	Confirms simulation artifacts are fit for RL training
	SIM	Begin contractor-led deformable bag modeling in Omniverse Newton engine	Engages external expertise for deformables modeling
	ORCHES.	First prototype of algorithm for heterogeneous fleets implemented in simulation	Unlocks algorithmic development under RMS architecture
Q2 2026	TACTILE	Long-duration PickMod experiment (≥20h) with humanoids under impedance/MPC control	Provides robust physical data for deformable bag manipulation; expands factory throughput from short-run to sustained operation
	TACTILE	Principal Research Scientist in RL hired and onboarded	Anchors RL and tactile workstreams with expert leadership
	SIM	Digital twin of BOS3 PickMod validated against TREAT Lab instrumentation	Calibration anchor for deformables modeling and benchmarking
	SIM	Initial exploration of WFM (Cosmos, GROOT-Dreams) applied to Chewy tasks	Establishes WFM capability/limits for benchmarking pipeline
	ORCHES.	Simulation benchmark of algorithm performance published internally	Enables decision on whether to continue or pivot
Q3 2026	TACTILE	Pilot tactile-augmented dataset of deformable bag manipulation published	Demonstrates tactile uplift over vision-only baselines; first external output from tactile data line of the factory
	TACTILE	RL-controlled humanoid grasp policy trained in simulation and validated in TREAT Lab	Shows Chewy datasets enhance manipulation performance
	SIM	Expanded deformables asset library released with calibration benchmarks	Confirms fidelity progress for deformables library
	SIM	Synthetic-Data Playbook draft completed with benchmarking results	Provides internal guidance on WFM use; formalizes integration between physical and digital factory lines
	ORCHES.	Prototype RMS algorithm validated in high-fidelity simulation	Demonstrates cross-line coordination; prepares orchestration line for end-to-end lab deployment
Q4 2026	TACTILE	Full-scale tactile dataset (≥200h) released across PickMod, pack, truck load	Establishes adoption of tactile dataset as industry standard; marks full-scale production milestone for tactile line
	TACTILE	Conference and journal papers submitted on dataset utility and tactile uplift	External credibility through peer-reviewed publications
	SIM	Library of validated deformable bag models published with artifacts	Provides reproducible deformables assets for industry; completes digital line's first full production cycle
	SIM	Synthetic-Data Playbook finalized and released publicly	Shapes industry use of WFMs in robotics pipelines
	ORCHES.	End-to-end RMS demo in TREAT Lab with ≥2 humanoid platforms + 1 mobile robot	Demonstrates heterogeneous fleet orchestration in FC-like environment; validates factory-scale coordination under RMS
	ORCHES.	Prototype RMS algorithm submitted for external publication	Establishes RMS as foundational control software for the data factory and future FC deployment
By year-end, the TREAT Lab operates as a coordinated data factory with three synchronized lines producing datasets, simulation assets, and orchestration algorithms that collectively define the standard for warehouse humanoids.			
Budget			

Chewy’s FY26 capital plan sustains the build-out and operation of the **TREAT Lab as a data factory**. The lab remains our core infrastructure for Physical AI, and maintaining it requires steady investment in humanoid platforms, sensing and compute systems, and the facility improvements that enable safe, continuous experimentation. These capital expenses complete the lab’s commissioning, expand its capability for multi-vendor testing, and ensure the data factory operates efficiently across tactile, simulation, and orchestration lines.

Our goal is to maximize throughput: more robots, richer data, faster iteration. The budget reflects this principle. We direct capital toward humanoid acquisition, sensor-rich experimentation, and the compute infrastructure required for large-scale training of physical AI models. Maintaining and improving the lab’s physical environment (its safety systems, fixtures, and instrumentation) ensures reliable, repeatable data collection.

OPEX	FY26	Description
Software	\$90k	Licenses for simulation, data labeling, and robotic control environments.
AWS	\$217k	Cloud compute and storage for large-scale model training and data capture.
Simulation Consulting	\$100k	External expertise for deformable modeling.
MassRobotics Sponsorship	\$80k	Annual partnership supporting industry collaboration, vendor access, and visibility within the robotics ecosystem.
Professional Memberships	\$10k	Organizational memberships in technical societies and consortia.
Robot Lease	\$80k	Short-term leases for evaluation of humanoids in the TREAT Lab.
Supplies, Quick-Turn Services	\$151k	Consumables, fabrication materials, and external machining or electronics services that keep lab operations running smoothly.
Travel & Entertainment	\$40k	Conference participation, vendor visits, and academic collaboration travel.
OPEX Total	\$768k	
CAPEX		
Humanoid Robots	\$450k	Purchase of humanoid research platforms across multiple vendors to build a heterogeneous fleet for evaluation and data generation.
Sensing and Instrumentation	\$350k	Tactile sensors, LiDAR, motion capture, depth and high-speed cameras, force/torque sensors, safety barriers, load-bearing platforms, and other lab instrumentation essential to operate the data factory safely and at scale.
Fabrication and Lab Infra	\$300k	fabrication tools, electronics benches, 3D printers and facility upgrades including fixtures, alignment systems, PickMod upgrades for long experiments, and electrical improvements required to maintain high-availability operation of the lab.
Compute & Networking	\$200k	On-premise compute and networking systems supporting physical AI training and coordination workloads. Complements AWS-based compute for large-scale model training and simulation.
CAPEX Total	\$1,300k	
TOTAL	\$2,068k	

Hiring

In 2026 we will add a Principal Research Scientist in reinforcement learning to anchor tactile and simulation efforts, strengthen the “middle layer” with engineers who can keep the data factory running at high throughput, and hire an AI Technologist to translate progress into clear internal visibility. We will also expand our academic pipeline by hosting graduate students for six-month sabbaticals focused on impactful research (e.g., RMS optimization or tactile-based grasping) and undergraduate interns who design and run experimental protocols, with Campus Recruiting supporting this new graduate initiative.

Full Time Headcount	2025	2026	Co-Op	2026
Allocated	13	15 (+2)	Grad	2
In-seat	9	15 (+6)	Undergrad	2

Level	Title	Skills	Notes
C08	Principal Research Scientist, Physical AI	Reinforcement learning to fine-tune physical AI models	New

C07	Staff Robotics Engineer, Physical AI	Support reinforcement learning	New
C07	Staff Robotics Engineer, Algorithms	Optimization, multi-agent support	Backfill
C06	Senior Test Engineer, Lab Owner	Instrumentation, lab operations, experiments owner	Backfill
C05	AI Technologist	Multi-modal AI models	Backfill
C04	Robotics Software Engineer I	Support teleop, data collection, controls	Backfill

Dependencies and Asks

Sustaining the data factory requires coordination across key internal functions, where early alignment ensures smooth operation and growing visibility.

Campus Recruiting. We are launching a graduate researcher program in 2026, bringing PhD students for six-month sabbaticals alongside an expanded undergraduate co-op program that will recruit from universities in the greater Boston area. All students must be able to work from the TREAT Lab in BOS3, and we ask Campus Recruiting to extend its processes to fully support co-ops and sabbaticals at this site, ensuring a steady pipeline of skilled operators and researchers to keep the data factory productive.

Communications. Our ability to influence depends on sharing results widely. We ask Communications to make the promotion of datasets, partnerships, and publications the default stance, presenting the TREAT Lab as a national proving ground for physical AI and highlighting how Chewy’s factory-scale approach is shaping the field.

Conclusion and Next Steps

Chewy’s Physical AI program is built to operate as a **data factory**—an integrated system that produces the datasets, benchmarks, and orchestration tools that will define humanoid performance in warehouse environments. In 2026, this factory begins sustained operation. Each robot, experiment, and model contributes to a cycle of production that strengthens Chewy’s influence on the field while advancing our readiness for humanoid deployment.

The factory’s operation depends on its production lines. The tactile line captures data that teach robots how to manipulate heavy, flexible items. The simulation line extends that work into digital space, generating synthetic data and validating digital twins. The orchestration line acts as the control system, managing how heterogeneous robots work together as a coordinated fleet. Each line feeds the others: physical trials refine simulations, simulations accelerate control development, and orchestration expands the scale and efficiency of testing. Together they form a self-reinforcing production system that grows more capable with every run.

Execution in 2026 demonstrates that system at work. The tactile line will release open datasets that improve deformable manipulation performance. The simulation line will publish validated digital assets and guidance on how to blend synthetic and physical data. The orchestration line will show multiple humanoids operating under a shared RMS, coordinated and benchmarked under Chewy’s standards. These outcomes prove that the data factory can operate continuously, safely, and at scale.

Risks remain in hardware maturity, sensor availability, and simulation fidelity, but none undermine the foundation we have built. The processes now in motion (data generation, simulation validation, and orchestration) can run independently and compound over time.

By year-end, success will be measured by steady production. Robots will be completing tasks, datasets will be generated and released, and orchestration algorithms will be managing mixed-vendor fleets. **2026 is the year Chewy’s data factory begins full production, establishing the systems, tools, and standards that will shape humanoid robotics for warehouse operations.**

Appendix

FAQs

1. Why not wait for humanoid robots to be available on the open market?

It might seem prudent to wait until humanoids are commercially proven before investing, but that approach would leave Chewy as a late adopter in a fast-forming market. The humanoid sector is moving quickly. One vendor recently announced the completion of a months-long trial in a BMW’s body-panel line, showing that the automaker is already shaping the robot to meet its specific manufacturing needs. Companies that engage early are defining what these robots will be capable of. If Chewy waits for the technology to mature, we risk buying robots designed for other customers’ workflows rather than our own. Each vendor is still making trade-offs in dexterity, strength, and mobility, and their training priorities will follow the data available to them.

By engaging the robotics community now, we can help accelerate the readiness of humanoids with the capabilities Chewy requires. Our open strategy—publishing datasets, developing orchestration systems, and hosting vendors in the TREAT Lab—gives us influence over how the field evolves. Vendors use our facilities and data to prove their robots on realistic FC tasks, while we learn from their progress without committing to exclusivity. This approach keeps Chewy visible to all major players and ensures that when humanoids reach commercial readiness, they arrive *pretrained on our work* and compatible with our RMS.

Bake-offs and proof-of-concept trials will follow when the technology matures and ROI thresholds are clear. Acting now positions Chewy to shape the industry’s trajectory rather than waiting for others to define it.

2. What is the readiness timeline for humanoid robots in FC operations?

Our base case is five years to achieve broad economic viability, with bookends considered at two and eight years.

- **Within two years:** This outcome is improbable. It would require robots to demonstrate narrow autonomy in simple, constrained workflows like truck loading or box pushing, with units capable of running 12–16 hours per day at costs near \$20–25 per productive hour. One vendor has recently reported a multi-month trial in an automaker’s body-panel line, which is a positive sign of durability. However, the task involved pushing buttons or lifting lightweight sheet metal in a fixed position, and the company has not claimed autonomous operation. These demonstrations indicate incremental progress in reliability, but they do not yet show the dexterity, perception, or adaptive control needed for warehouse environments that handle deformable products. If true readiness were near, we would already see credible pilots in unstructured settings, OEMs shipping hundreds of durable units, or vendors publishing reliability metrics and safety certifications. Those signals remain absent, which makes the two-year scenario unlikely.
- **Around five years:** This is our most plausible scenario. A realistic sequence looks like this: within two years, Physical AI models show deformable lifting in controlled lab environments; hardware matures in parallel, with actuators, power systems, and hands becoming more reliable; vendors begin advertising increased factory capacity and publish mean-time-between-failure metrics; and OEMs prove robots can work reliably for months in pilot deployments. By year five, costs fall toward \$15-22 per productive hour, and robots are credible for deployment at fleet scale in FC operations. This scenario aligns with our planning horizon.
- **Eight years or more:** Delays are possible. Like autonomous vehicles, humanoids face “long-tail” safety risks (dropped objects, unexpected collisions, recovery in mixed human environments) that must be validated over millions of operational hours. Regulators may withhold approval until such evidence is accumulated. In this scenario, broad deployment could slip into the 2030s, with incremental automation filling gaps in the meantime.

Chewy’s strategy is resilient across these horizons. By publishing datasets and developing orchestration systems now, we shape vendor roadmaps and ensure that when humanoids are ready—whether in two, five, or eight years—they are trained on our work and integrate smoothly into our fulfillment centers.

3. Influence through data publication seems wishful, how will the impact of our activities be measured?

A primary measure of success will be the partnerships we attract. By keeping our datasets and lab environments open and nonexclusive, we create a platform that vendors and academics want to use. Humanoid companies will bring their robots into the TREAT Lab to generate evidence of capability in realistic warehouse environments, and academic groups will collaborate with us to publish new methods on deformable manipulation and fleet orchestration. The number of vendors contributing data and the number of academic collaborations we support are direct signals that our openness is working.

We will also track both leading and lagging indicators. Lagging indicators include citations in peer-reviewed publications, dataset downloads on platforms like Hugging Face and GitHub, and letters of support from academic and industrial partners. Leading indicators include evidence that vendors incorporate our datasets into their training pipelines, measurable improvements in robot performance when models are fine-tuned on our data, and inclusion of our digital twins or RMS benchmarks in research challenges. These measurements provide a clear picture of how our work shapes the field.

4. Giving away our data is a one-way door. Are we sure this is the right approach?

Yes. The value of publishing is that it ensures every vendor trains their robots on *our* tasks. Physical AI models today are trained on only a few thousand hours of demonstrations, far less than the trillions of tokens GPT-4 required. This imbalance means that whoever provides rich, task-relevant datasets will disproportionately shape the capabilities of the coming wave of humanoid workers. Our aim is that all humanoids become skilled in lifting heavy deformables, providing Chewy optionality in future vendor selection.

The data we publish are not sensitive to Chewy’s operations. It consists of low-level measurements of robots and humans performing FC-like tasks, not business processes or proprietary information. Once released, these datasets cannot be clawed back—but that is the point. Exclusivity would limit adoption, while open publication maximizes it, shaping vendor roadmaps without locking us into a single partner.

The one-way door here is intentional. By making our data indispensable, we ensure humanoid robots arrive pretrained on Chewy’s work, giving us influence and flexibility when the market matures.

Glossary

Dataset Publication Peer-reviewed venues such as academic conferences and journals where research is formally reviewed and published. Examples include the *IEEE International Conference on Robotics and Automation (ICRA)* and *NeurIPS*. Publishing in these venues builds credibility and shapes technical roadmaps across the robotics industry.

Deformable Assets Library A collection of digital simulation assets representing flexible or soft objects commonly handled in Chewy’s fulfillment centers—bags of pet food, cat litter, dog beds, and similar items. These models replicate physical properties such as squishiness, surface tackiness, and particle sloshing, allowing robots to learn how such items behave when lifted, squeezed, or moved.

Synthetic Data Playbook An internal guide that outlines the sequence of steps engineers follow to integrate and validate new World Foundation Models (WFMs) in Chewy’s data-generation workflows. The Playbook captures lessons learned and establishes repeatable practices for future simulation projects.

Physical AI Models Multi-modal machine learning models designed to control robots. These models take in information from vision (camera data), natural language (instructions), and proprioception (the robot’s sense of its own position and motion). They output “tokens” that guide how the robot acts in the physical world, much like large language models generate words.

Tactile Sensors Contact sensors that measure pressure and touch across a surface with fine spatial resolution. Advances in flexible circuits, materials, and edge computing now make these sensors viable for robotic hands, giving robots a sense of “touch” to improve grasping and manipulation.

- Robot Orchestration / Robot Management System (RMS)** Software that manages how multiple robots work together as a coordinated fleet. The RMS includes algorithms for task assignment (e.g., “pick item A and place it in carton B”), path finding (navigation), fleet safety, maintenance scheduling, and energy management. It serves as a layer between Chewy’s Warehouse Management System (which assigns work) and the individual robots that perform it.
- Warehouse Management System (WMS)** The existing system that assigns work to people in fulfillment centers. WMS rules are written for human comprehension and execution.
- World Foundation Models (WFMs)** A new class of AI-based simulation environments that can generate synthetic, physics-aware training data for robots. WFMs use generative AI to create realistic virtual “worlds” where robots can practice tasks that would otherwise require costly physical experiments. Examples include NVIDIA’s *Cosmos* and *GROOT-Dreams*.
- Digital Twin** A virtual replica of a physical system such as a robot, workstation, or fulfillment cell. Digital twins allow engineers to test control policies and perform simulation-based experiments before deploying to real hardware.
- Sim-to-Real Gap** The performance difference between a robot’s behavior in simulation and its performance in the physical world. Closing this gap ensures that models trained in simulation can operate reliably in real environments.
- Large-Scale Task Assignment (LSTA)** A class of optimization algorithms used by the RMS to allocate many tasks across a fleet of robots. LSTA considers factors such as robot location, workload, charge level, and priority of orders to maximize throughput.
- Multi-Agent Path Finding (MAPF)** Algorithms that plan safe and efficient routes for multiple robots moving simultaneously within the same environment. MAPF ensures that robots avoid collisions and congestion while completing tasks.
- Federated Learning** A distributed learning technique where multiple robots share improvements to their control policies without sending all their raw data to a central server. It allows a fleet to learn collectively while maintaining data efficiency.
- TREAT Lab** Chewy’s Testing and Research of Embodied AI Technologies Lab at BOS3, a fully instrumented facility that replicates key fulfillment center processes such as picking, packing, and truck loading. The lab provides the environment for controlled experiments with humans and humanoids.
- Domain Randomization** A simulation technique that varies environmental factors—lighting, friction, object positions, and other details—so that models trained in simulation learn to focus on general principles rather than specific configurations. This helps models transfer successfully to real-world conditions.
- Deformable Object** An object whose shape changes when handled, such as a flexible bag or cushion. Deformable modeling is among the most difficult problems in robotic manipulation because it requires accurate physics to predict how material will shift and respond to force.
- Fleet Heterogeneity** The condition of operating robots from multiple vendors, each with different capabilities, dexterities, or interfaces. Chewy’s RMS is designed to coordinate such mixed fleets rather than rely on a single vendor’s system.

Expanded Rationale for 2026 Research Themes

Sense and Sensibility (data publication + tactile sensing)

Our central bet continues to be the publication of open datasets from humans and humanoids performing the manipulation tasks that matter most in Chewy FCs: picking, packing, and truck loading. In 2026, we will extend the work begun in 2025 by recording synchronized streams of human and robot trials in the TREAT Lab, curating them into formats fit for industry consumption, and publishing results that demonstrate their utility for training physical AI models. We will validate this utility by fine-tuning open-weight physical AI models on our processes and documenting the performance improvements. This establishes Chewy’s datasets as indispensable training resources and reinforces our strategy of making the world’s robots arrive pretrained on our work. Publication of these results in leading venues will demonstrate both the credibility of our science and the influence of our data.

What is new in 2026 is the addition of tactile sensing as a modality. For decades, researchers attempted to add tactile sensors to robot grippers to detect slippage and stabilize grasps. The idea showed promise but never scaled, because tactile sensors produce many simultaneous signals across a hand while classic controllers were designed to handle only a few low-order inputs. The mismatch made it infeasible to translate tactile input into meaningful control. With modern AI, this bottleneck is removed. Physical AI models are designed to interpret high-dimensional, multimodal data in the same way they already extract semantic meaning from images, natural language, and proprioception. This makes 2026 the right moment to invest, because the modality that was once too complicated to distill by human engineers can now become meaningful to AI.

Analog Devices, a long-established semiconductor manufacturer with deep expertise in MEMS sensors and a significant Boston-area research presence, is developing prototype tactile devices that fit this need. By collaborating with Analog Devices to integrate tactile sensors into humanoid platforms, we will produce the world's first open dataset of two-handed tactile manipulation of deformable bags. While tactile sensing is not the new center of gravity for our program, it is an important extension that will pave the road to publishing and expand the scientific relevance of our datasets. It also creates new opportunities: demonstrating how tactile feedback improves model fine-tuning, motivating the need for reinforcement learning expertise within our team, and signaling to the robotics community that Chewy is pushing the frontier of multimodal physical AI. Success will be measured by publications and by adoption of our tactile-augmented datasets as training resources by vendors and academics.

The World Is Not Enough (simulation + WFMs)

Our simulation program remains the backbone of Chewy's Physical AI strategy. In 2025 we committed to building digital twins of our robots and fulfillment center environments, developing models of deformable bags, and curating datasets suitable for reinforcement learning. That work continues in 2026 with greater emphasis on fidelity where it matters. Deformable bags remain a strategic priority: they represent the highest-volume, most physically challenging items in our FCs, and solving their manipulation is critical to unlocking humanoid utility in picking, packing, and truck loading. By extending our deformables library, validating digital twins of robots and PickMods, and refining domain randomization techniques, we will ensure our datasets transfer effectively from simulation to the TREAT Lab. Success will be measured by reduced sim-to-real performance gaps and faster controller bring-up cycles, demonstrating to the broader field that Chewy's simulation artifacts are both credible and indispensable.

What is new in 2026 is our deliberate exploration of World Foundation Models (WFMs). Recent AI-based simulation tools can generate physics-aware robot episodes from simple inputs, reducing the need for months of manual data collection. WFMs are both threat and opportunity: if ignored, they risk diluting the influence of the curated datasets we have labored to produce; if leveraged well, they could become force multipliers. In 2026 we will benchmark WFM-only training against WFM-plus-Chewy datasets, publish results that clarify the additive value of real-world deformables data, and create a Synthetic-Data Playbook to guide internal use. By combining WFMs with high-quality assets, digital twins, and domain randomization, we aim to shape how the field integrates synthetic and experimental data and ensure Chewy's role remains central even as the simulation landscape evolves.

Band of Robots (RMS and heterogeneous orchestration)

In 2025, we defined the Robot Management System (RMS) architecture as a necessary complement to the Warehouse Management System (WMS), responsible for orchestrating humanoids as a cohesive fleet rather than as isolated agents. The business case extends beyond internal efficiency. An RMS that is hardened in Chewy's operational environment could evolve into a lucrative standalone product, addressing a nascent market for heterogeneous humanoid orchestration. Today's WMS assigns work to people using simple, rule-based instructions designed for a large hourly workforce. An RMS performs the same coordinating function for robots, with all the advantages of a mechanized fleet including instantaneous communication, shared learning, and system-wide optimization. By owning this orchestration layer, we prepare Chewy's FCs for humanoids while also creating the foundation for a potential new business line that no humanoid vendor is positioned to deliver.

That work continues in 2026 with an emphasis on algorithms and prototypes. Our focus areas are large-scale task assignment, multi-agent path finding, collaborative localization, and fleet-level safety. The TREAT Lab will serve as the proving ground where algorithms tested in simulation are exercised on real robots. By releasing open benchmarks that

528 reflect Chewy's FC topologies, we will fill a gap in the research community and establish Chewy as a leader for
529 heterogeneous fleet orchestration. Success will be measured by reduced congestion in FC-like environments, improved
530 throughput over single-agent baselines, and external uptake of our RMS benchmarks.
531

532 What is new in 2026 is our explicit embrace of heterogeneity. Chewy will never procure humanoids exclusively from a
533 single vendor; differences in mobility, dexterity, payload, and software interfaces make such a strategy impractical. Our
534 RMS must orchestrate across these differences, ensuring that robots with varied capabilities work together as a coherent
535 fleet. This requires heterogeneity-aware task assignment algorithms, extensions to routing methods that account for
536 different edge costs by robot type, and federated learning to propagate skills across diverse embodiments. By year-end,
537 we aim to demonstrate an end-to-end RMS in the TREAT Lab coordinating at least two humanoid platforms alongside a
538 mobile robot, proving that heterogeneous fleets can be made productive through orchestration.
539